

# Dexter Fernandes

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<https://github.com/Dexter-Fernandes>

## Computer Vision Engineer

Computer Vision Engineer with 3+ years of experience delivering low-latency vision solutions for Tier 1 companies, including a construction firm in Texas and a UK oil and gas company. Proven track record of collaborating with multinational clients to deploy AI-driven vision solutions in fast-paced dynamic environments. Experienced in applying state-of-the-art AI research while leading testing, QA, and monitoring processes.

## WORK EXPERIENCE

Esbaar • Muscat, Oman

Jan 2023 - Aug 2023

### AI Engineer

- Developed time-series machine learning models for the classification of on-shore oil rig operations to increase worker safety and reduce supervision required during 24/7 operation, achieving a 23% higher F1 score than human experts.
- Built an automated sensor-data cleansing pipeline handling 3.6 million records, reducing model rollout time from 3 days to under 24 hours.
- Built automated **training and validation pipelines** integrated with **MLflow** for experiment tracking and **Optuna** for Bayesian hyperparameter tuning, cutting model iteration time by **40%**.
- Developed real-time perception modules for oil-rig safety systems, deploying CNN-based vision pipelines with YOLOv5 on edge devices to improve detection accuracy by 15%.
- Refactored legacy monolith into Kafka-driven microservices, containerised with Docker for fault isolation and 35% higher MTBF.
- Designed a dynamic memory management solution to address recurring crashes in a live video streaming system, reducing crash frequency by 67%.
- Enhanced code quality and documentation standards by leading formal peer code reviews and quality-assurance activities within each sprint cycle, ensuring best practices.
- Designed automated data collection, augmentation, and labeling tools to scale large image datasets, increasing dataset diversity by 40% and cutting manual labeling time by 60%.
- Oversaw on-site deployment of the system at a British Petroleum oil rig in the Omani desert.
- Implemented continuous integration and testing workflows for edge inference modules, improving maintainability and deployment speed across hardware targets.
- Collaborated with hardware teams and the client to validate software performance in real-world field tests and enhance system robustness.
- Integrated AI modules with production systems (Docker, FastAPI, Kafka, Git CI/CD) in collaboration with software/hardware/QA teams to deliver fault-tolerant edge deployments.
- Engineered production-grade AI software for real-time perception and safety analytics, delivering deterministic performance across 24/7 oil rig operations.
- Validated deterministic model performance under extreme field conditions at a British Petroleum site, ensuring reliability across 24/7 operations.
- Applied Agile and Scrum methodologies to plan sprints, manage deliverables, and ensure iterative software improvement across multidisciplinary teams.

LivNSense Technologies Pvt. Ltd. • Bengaluru, India

May 2020 - Dec 2022

### Lead Software Engineer - AI • Full-time

- Led a 4-member team to design, train, and deploy CNN-based models for real-time object detection, pose estimation, and tracking, reducing false positives by 33% and achieving sub-50 ms inference latency on edge hardware.

- Built and optimised deep learning pipelines with YOLOv5 and DeepSORT, enabling consistent object tracking and robust analytics in dynamic industrial environments.
- Deployed quantised OpenVINO vision models over Microsoft Azure to rugged Intel VPU edge devices for on-site inference in harsh industrial environments.
- Re-architected the embedded inference pipeline and model optimisation workflow, increasing latency from 5 FPS to 15 FPS, enabling real-time detection of safety-critical events.
- Repurposed legacy object detection models for automated labelling pipeline workflows, reducing manual annotation effort by 80% and accelerating dataset curation.
- Developed synthetic data generation and augmentation tools to improve edge-case coverage and cut manual labelling time by 60%.
- Developed core perception algorithms on resource-constrained Intel edge devices, leveraging mixed-precision and fixed-point quantization with structured pruning to meet strict memory and latency budgets for real-time decision-making in industrial environments.
- Led a 4-engineer team to design, train, and deploy CNN-based object detection and tracking models for industrial safety applications, reducing false positives by 33% and achieving sub-50 ms inference latency on edge devices.
- Optimised neural network architectures using quantisation-aware training, structured pruning, and layer fusion reducing model size by 50%.
- Implemented rigorous testing and validation frameworks simulating challenging lighting, occlusion, and motion conditions, improving CNN stability and reducing variance by 25%.
- Deployed a unified quantised CNN model for IR and RGB sensor streams using Intel OpenVINO, ensuring deterministic real-time inference on constrained edge devices.
- Collaborated with software, hardware and QA teams to integrate AI modules into **Docker/FastAPI** production systems using **Git CI/CD**, ensuring reproducible, fault-tolerant deployments.
- Engineered production-grade edge perception software for safety-critical environments, delivering deterministic real-time behaviour on Intel VPU devices.
- Mentored engineers on optimisation and deployment practices, raising code quality and delivery consistency.
- Presented key findings and metrics to leadership and clients, translating complex performance results into actionable insights for deployment and further optimisation.

## EDUCATION

### Master of Science in Robotics - Merit

University of Bristol • Bristol, England

Sep 2023 - Sep 2024

#### Radiological SLAM with LiDAR Odometry

- Designed and implemented a real-time SLAM system integrating LiDAR and RGB data for teleoperation of an Unmanned Ground Vehicle (UGV).
- Processed and analysed large-scale point cloud datasets using Open3D and RTAB-Map to perform 3D feature extraction, registration, and map optimization.
- Tuned SLAM parameters to improve localisation accuracy across varied terrains, achieving stable performance in both feature-dense indoor and feature-sparse outdoor environments.
- Simulated and validated 2D/3D mapping and navigation workflows in Gazebo and ROS 2 before successful real-world deployment.
- Integrated multiple sensing modalities to achieve robust spatial and radiological understanding, enabling accurate obstacle detection in static environments.
- Implemented waypoint navigation in simulation using ROS2 Nav2 (standard planners/controllers) within Gazebo, validating end-to-end navigation workflows.

## Bachelor of Technology in Electrical and Electronics Engineering

Visvesvaraya National Institute of Technology • Nagpur, India

Jul 2016 - May 2020

- Closed-loop oven temperature control: Implemented P/PI/PID strategies driving a 4–20 mA thyristor power unit to modulate heater output; calibrated analog I/O, added exhaust-fan load disturbances to test robustness, and tuned gains to maintain setpoint within  $\pm 2^\circ\text{C}$  during transients.
- DC motor speed control via semi-controlled thyristor bridge (0.5 HP shunt motor, 24 V encoder, PID): Wrote Ladder/FBD logic to generate firing-angle commands, scaled encoder feedback, and tuned PID gains to hold setpoint speed under load changes while reducing overshoot and steady-state error.
- Automated bottling station (dispense + rotary table): Programmed sequence control for an inductive alignment sensor, capacitive level sensor, solenoid fill valve, and stepper-driven turntable.

## CERTIFICATIONS

### Deep Learning Specialisation

Jul 2019

deeplearning.ai

## AWARDS & SCHOLARSHIPS

### Best Speaker award

Jan 2020

ABC Toastmasters club

## PROJECTS

### ROS2 SLAM & Nav2 Simulation

Oct 2025

- Built a simulated mobile robot integrating 2D LiDAR and RGB camera data in Gazebo Harmonic to perform real-time SLAM and autonomous navigation using ROS 2 Jazzy.
- Applied slam\_toolbox and Nav2 for 2D/3D mapping, localisation, and waypoint-based navigation with optimised Ceres backends.
- Processed multi-sensor data streams for spatial alignment and tested mapping accuracy across diverse simulated environments.
- Developed modular ROS 2 launch files and parameterised configurations for flexible experimentation with perception and navigation pipelines.
- Containerised the full system using Docker to ensure reproducibility, version control, and cross-platform deployment.

## SKILLS

- **Perception & SLAM:** 3D Feature Extraction, Data Augmentation, Denoising, LiDAR SLAM, NumPy, Open3D, OpenCV, Point Cloud Library, SAM, Scan Matching, Scikit-Image, U-Net, Vision Transformer (ViT), YOLO
- **Cloud, Edge & MLOps:** Apache Kafka, AWS, Docker, DVC (Data Version Control), FastAPI, Git, Intel VPUs, Linux, Microsoft Azure, MLFlow, NVIDIA Jetson
- **Programming & Development:** Bash, C++, CUDA, Django, Gazebo (Harmonic), Nav2, Pandas, PostgreSQL, Python, Qt, ROS1 (Melodic/Noetic), ROS2 (Humble/Foxy/Jazzy)
- **Deep Learning and Optimisation:** Bayesian Optimisation, Convolutional Neural Networks, DeepSORT, MultiRocket, ONNX, OpenVINO, Optuna, PyTorch, Quantisation-Aware Training, TensorFlow, TensorRT